

	ϵ	0	1	00	01	11	000	001	--
M_ϵ	H	L	H	H	L	L	L	H	--
M_0	L	H	L	L	H	L	H	H	
M_1	H	L	H	H	L	H	L	H	
M_{00}	L	L	L	H	L	H	H	H	
M_{01}	H	H	L	H	H	L	L	L	
M_{11}	L	L	H	H	H	L	H	L	
M_{000}	L	H	L	H	L	H	H	L	
M_{001}	H	L	H	H	L	H	H	H	

Suppose $\exists$ a total TM K such that $L(K) = HP$

For any x and y , K can determine the entry in the $(x,y)^{th}$ cell in the above table.

On input $M\#x$,

- K halts and accepts if M halts on x
- K halts and rejects if M loops on x

Consider a TM N that on input $x \in \{0,1\}^*$ does the following:

- 1) Constructs M_x from x and writes $M_x\#x$ on its tape
- 2) Runs K on input $M_x\#x$, accepting if K rejects and going into a trivial loop if K accepts.

For any $x \in \{0,1\}^*$, N halts on x iff K rejects $M_x\#x$
iff M_x loops on x .

That is, N is different from every M_x on at least one string - the string x .
This gives a contradiction.

$HP = \{ M \# x \mid M \text{ halts on } x \}$ is not recursive

HP is recursively enumerable.

Membership Problem.

$MP = \{ M \# x \mid x \in L(M) \}$.

MP is r.e

Question. IS MP recursive?

$$HP = \{M\#x \mid M \text{ halts on } x\}$$

$$MP = \{M\#x \mid x \in L(M)\}.$$

Theorem. MP is not recursive.

Suppose $\exists$ a total TM K s.t. $L(K) = MP$.

We can use K as a sub-routine to solve the halting problem.

That is, can construct a total TM K' (using K) s.t.

$$L(K') = HP \quad (\text{this is a contradiction})$$

Theorem. MP is not recursive.

Suppose $\exists$ a total TM K s.t. $L(K) = MP$.

Given a TM M and input x . To check if M halts on x

Build a new TM N_M that does the following.

- Similar to M , N_M accepts if M accepts or rejects.

For all $x \in \Sigma^*$, N_M accepts x iff M halts on x .

Theorem. MP is not recursive.

Suppose $\exists$ a total TM K s.t. $L(K) = MP$.

Given a TM M and input x . To check if M halts on x
Build a new TM N_M that does the following.

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For all $x \in \Sigma^*$, N_M accepts x iff M halts on x .

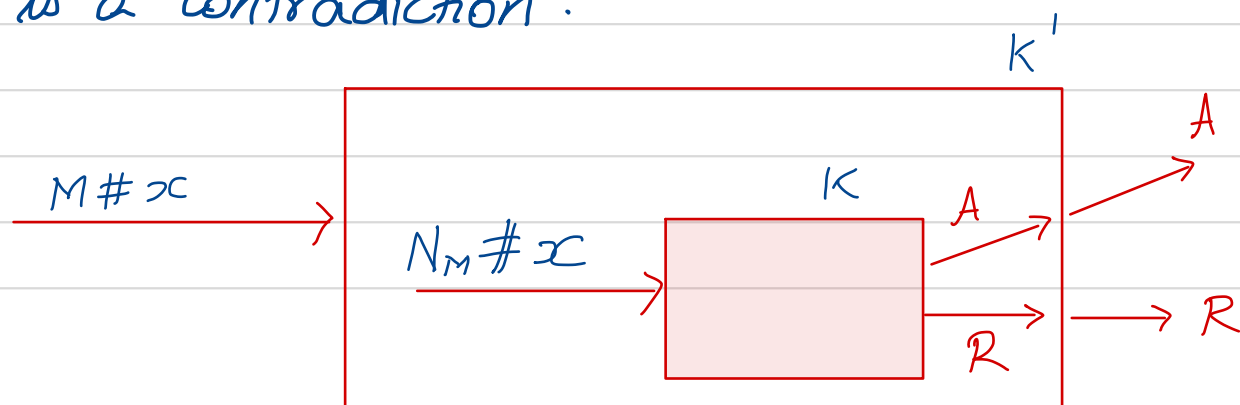
For any m & x , to check if m halts on x

Construct N_m and run K on input $N_m \# x$

By assumption K is a total TM. But then we

can construct a total TM K' s.t. $L(K') = HP$

This is a contradiction.



HP is not recursive, HP is r.e

MP is not recursive, MP is r.e

Question. Is $\overline{HP}$ r.e.?

Question. Is $\overline{MP}$ r.e.?

Lemma. if A is r.e and $\overline{A}$ is r.e then A is recursive

Therefore

- $\overline{HP}$ is not r.e

- $\overline{MP}$ is not r.e.